

PRIVACYSHIELD

Digital Footprint & Privacy Risk Auditor

"See Everything. Control Everything. Own Your Digital Life."

WEB APP

MOBILE APP

AI-POWERED

BLOCKCHAIN

CSI TSEC Hackathon 4.0 | 2026

Problem Statement: Digital Footprint & Privacy Risk Auditor

THE PROBLEM

People have dozens of accounts, reused passwords, and apps with broad permissions — yet have **no idea how exposed they are**. Accounts are quietly linked through email recovery and SSO, so **one compromised account can open the door to many others**. Existing tools check a single password or breach but never show how everything connects or what to fix first.

OUR APPROACH

PrivacyShield is a **cross-platform (Web + Mobile) AI-powered privacy command center** backed by **blockchain-verified audit trails**. It maps your entire digital footprint into an interactive graph, uses AI to score cascading risk across account chains, and stores every audit/fix action on-chain so your privacy posture is provably immutable and tamper-proof.

WHY AI + BLOCKCHAIN?

- AI: Graph neural networks model cascading risk across interconnected accounts — not just isolated checks.
- AI: NLP scans privacy policies & T&C; changes in real-time, alerting you to hidden data-sharing clauses.
- AI: Predictive breach probability engine — knows which of YOUR accounts is most likely to be hit next.
- Blockchain: Every risk score, audit, and fix action is hashed on-chain — immutable proof of your privacy posture.
- Blockchain: Zero-Knowledge Proofs let you prove your security level to insurers/employers without revealing account details.
- Blockchain: Decentralized identity (DID) replaces centralized SSO — you own your login, not Google/Apple.

CORE FEATURES

[1] Accounts & Apps Inventory (Web + App)

A unified vault to catalog every digital account, app permission, and connection — without storing real passwords.

- **Smart Import:** Auto-detect accounts from email inbox scanning (OAuth read-only), browser extensions, and app permission APIs
- **Manual Add:** Add accounts with service name, email used, 2FA status, password-reuse group label, recovery methods
- **App Permission Scanner (Mobile):** Native integration with Android/iOS permission APIs to list every app's access (camera, location, contacts, mic, storage)
- **SSO & Recovery Chain Mapping:** Map which accounts use Google/Apple/Facebook SSO, and which emails serve as recovery for others
- **Password Reuse Groups:** Label accounts sharing the same password (no real passwords stored) — see exactly how many doors one leaked password opens
- **Blockchain Anchor:** Hash of your full inventory stored on-chain as a timestamped snapshot — proves your audit state at any point in time

[2] Exposure Mapping & Risk Scoring (AI-Powered Core Challenge)

The brain of PrivacyShield — a graph-based AI engine that models cascading risk across your entire digital network.

- **Interactive Account Graph:** Visual force-directed graph showing every account as a node, with edges for SSO links, recovery chains, shared passwords, and data-sharing connections
- **Cascading Risk Engine (GNN):** Graph Neural Network propagates risk through connections — a weak email account that can reset 10 others gets a critical score even if it has 2FA
- **Breach History Integration:** Auto-check accounts against HaveIBeenPwned and breach databases; flag actively compromised credentials
- **Single Point of Failure Detection:** AI identifies "keystone" accounts — the one email/phone that if compromised unlocks everything
- **Risk Score Formula:** Composite score per account: breach history (30%) + permission scope (20%) + password reuse (20%) + missing 2FA (15%) + cascading impact (15%)
- **Network-Wide Privacy Score:** Overall 0-100 score reflecting your entire digital footprint health, weighted by cascading dependencies

[3] AI Fix Checklist & Guided Remediation

Not just alerts — actionable, prioritized steps ranked by how much overall risk each one removes.

- **Priority-Ranked Actions:** AI ranks fixes by network-wide risk reduction — enabling 2FA on your primary Gmail might remove 40 points of cascading risk across 15 accounts
- **One-Tap Fix Links:** Deep-links to each service's security settings page (2FA setup, password change, permission revoke, account deletion)
- **Guided Walkthroughs:** Step-by-step visual guides for non-technical users (e.g., 'How to enable 2FA on Instagram')

- **Fix Impact Preview:** Before and after risk score simulation — see how much your score improves BEFORE doing the fix
- **Progress Tracking:** Checklist tracks completed vs. pending fixes with completion percentage and estimated time
- **Blockchain Fix Receipt:** Each completed fix action is hashed on-chain — immutable proof you took the action at that timestamp

CORE FEATURES (CONTINUED)

[4] Search, Filters & Exploration

Navigate your digital footprint with powerful search and drill-down.

- **Global Search:** Instant search across accounts, services, permissions, risk levels, and fix actions
- **Smart Filters:** Filter by risk level (Critical/High/Medium/Low), service type (Social/Finance/Email/Cloud), 2FA status, data shared, last activity date
- **Graph Filters:** Highlight specific connection types on the graph — show only SSO chains, only password-reuse clusters, or only recovery links
- **Timeline View:** See when each account was added, last audited, and breach history chronologically

[5] Reminders, Breach Alerts & Continuous Monitoring

Always-on vigilance — never be blindsided by a breach again.

- **Real-Time Breach Monitor:** Continuous background scanning against breach databases; instant push notification (web + mobile) when your data appears in a new breach
- **Cascade Alert:** When one service is breached, AI instantly maps which linked accounts are now at risk and generates an emergency fix checklist
- **Periodic Review Reminders:** Scheduled prompts (weekly/monthly) to review permissions, update passwords, and re-audit
- **Dark Web Scanning (AI):** NLP-powered scanning of dark web paste sites and forums for your email/username mentions
- **Smart Notification Priority:** AI categorizes alerts as Critical (act now), Warning (act this week), Info (awareness only) — no alert fatigue

[6] Privacy Dashboard — The Command Center

One view. Full visibility. Every metric that matters.

- **Overall Privacy Score:** Large 0-100 gauge with trend arrow showing improvement/decline over time
- **Risk Heatmap:** Accounts colored by risk level on the interactive graph — red clusters scream for attention
- **Single Points of Failure:** Top 3 keystone accounts highlighted with their cascading impact radius
- **Fix Progress:** Completion ring showing % of recommended fixes done, with estimated time remaining
- **Connection Graph Mini-Map:** Zoomable mini-view of your full account network with edge coloring by connection type
- **Improvement Timeline:** Historical chart showing your privacy score over weeks/months as you complete fixes
- **Breach Exposure Summary:** Count of breaches affecting your accounts, total records exposed, and remediation status

AI FEATURES DEEP DIVE

[AI-1] Privacy Policy Watchdog (NLP Engine)

AI reads the fine print so you don't have to.

- **Auto-Scan T&Cs:** NLP model ingests Terms of Service and Privacy Policies of your connected services
- **Plain-English Summaries:** Converts legalese into simple risk summaries — 'This app shares your location data with 47 third-party advertisers'
- **Change Detection:** Monitors policy updates and alerts you when a service quietly adds new data-sharing clauses
- **Risk Flags:** Auto-flags clauses about data selling, indefinite retention, cross-border transfers, and government disclosure

[AI-2] Predictive Breach Probability

Which of YOUR accounts is most likely to be breached next?

- **Historical Pattern Analysis:** ML model trained on breach frequency, company size, sector vulnerability patterns, and security posture signals
- **Per-Account Probability:** Each account gets a predicted breach probability score for the next 6 months
- **Proactive Hardening:** AI recommends preemptive actions for high-probability targets BEFORE a breach occurs
- **Sector Risk Trends:** Dashboard showing which sectors (healthcare, fintech, social media) are trending toward higher breach rates

[AI-3] AI Chat Assistant — PrivacyBot

Ask anything about your digital security in natural language.

- **Natural Language Queries:** 'Which of my accounts are most at risk?' / 'What happens if my Gmail gets hacked?' / 'How do I delete my old Facebook?'
- **Scenario Simulation:** 'What if my phone is stolen?' — AI simulates the attack chain through your account graph and shows exposure
- **Personalized Advice:** Context-aware recommendations based on YOUR specific account network, not generic tips
- **Voice Input (Mobile):** Ask questions via voice on the mobile app for hands-free security checks

[AI-4] Smart Permission Advisor

AI analyzes whether each app actually NEEDS the permissions it has.

- **Permission vs. Usage Analysis:** Cross-reference granted permissions with app category — a calculator app with microphone access gets flagged
- **Risk-per-Permission Score:** Quantify how much risk each permission adds (location + contacts + camera = high surveillance potential)
- **Bulk Revoke Suggestions:** One-tap list of permissions you can safely revoke without breaking app functionality
- **App Alternatives:** AI suggests privacy-respecting alternatives for high-risk apps (e.g., Signal instead of a messaging app with poor privacy)

BLOCKCHAIN FEATURES DEEP DIVE

[BC-1] Immutable Audit Trail

Every privacy action you take is permanently and verifiably recorded.

- **On-Chain Hashing:** SHA-256 hash of every audit snapshot, risk score change, and fix action stored on Polygon/Ethereum L2
- **Timestamped Proof:** Prove to any third party (insurer, employer, regulator) exactly when you audited your accounts and what actions you took
- **Tamper-Proof History:** No one — not even you — can retroactively alter your privacy audit history
- **Gas-Efficient:** Batch multiple actions into Merkle tree roots — one on-chain transaction covers hundreds of actions

[BC-2] Zero-Knowledge Privacy Proofs (ZKP)

Prove your security level WITHOUT revealing your account details.

- **ZK-SNARK Certificates:** Generate a cryptographic proof that your privacy score is above a threshold (e.g., >80) without revealing which accounts you have or their individual scores
- **Use Cases:** Share ZKP with cyber insurance providers for lower premiums, with employers for compliance, or with platforms for trust verification
- **Selective Disclosure:** Choose exactly what to prove — 'I have 2FA on all financial accounts' without listing which financial accounts you use
- **Verifiable On-Chain:** Any verifier can check the proof against the blockchain without accessing your data

[BC-3] Decentralized Identity (DID) Integration

Own your identity — stop depending on Google/Apple/Facebook for login.

- **Self-Sovereign Identity:** Create a DID (W3C standard) anchored on blockchain — your login credential that no company can revoke
- **Replace SSO Dependency:** Gradually migrate from Google SSO to DID-based authentication, eliminating single-provider failure
- **Portable Reputation:** Your privacy score and audit history travel with your DID across platforms
- **Wallet Integration:** MetaMask / WalletConnect for blockchain interactions; no crypto knowledge required for basic features

[BC-4] Decentralized Breach Reporting (DAO)

Community-powered breach intelligence that can't be censored or delayed.

- **Breach Report DAO:** Anyone can submit a breach report to the decentralized network; community validators verify before alerting users
- **Incentivized Reporting:** Token rewards for verified breach reports — security researchers earn tokens for early disclosures
- **Faster Than Official:** Community-reported breaches often surface days/weeks before companies officially disclose
- **Censorship-Resistant:** No single entity can suppress or delay breach notifications

CRAZY EXTRA FEATURES

Beyond the requirements — features that make PrivacyShield unforgettable at the hackathon.

[X1] "HACK ME" Attack Simulator

Simulate a real attack on YOUR account network to see exactly what an attacker could reach.

- **Choose Attack Entry Point:** Pick any account and simulate it being compromised
- **Animated Attack Chain:** Watch the attack propagate through your graph in real-time — SSO links, recovery chains, password reuse — each hop lights up red
- **Damage Report:** After simulation, see total accounts reachable, data exposed, and financial accounts at risk
- **Before/After:** Run the simulation again AFTER applying recommended fixes to see the attack chain shrink

[X2] Privacy Score NFT Badge

Mint your privacy score as a dynamic, on-chain NFT that updates as your score changes.

- **Dynamic NFT:** SVG-based NFT that visually changes color/design based on your current privacy score (green shield = strong, red skull = weak)
- **Flex Your Security:** Display on social profiles, resumes, or portfolios — verifiable proof of your digital hygiene
- **Leaderboard:** Anonymous global leaderboard of privacy scores — compete to be the most secure
- **Achievement Badges:** Earn special NFTs for milestones — '2FA Everywhere', 'Zero Reused Passwords', 'Dark Web Clean'

[X3] "Digital Death Switch" — Emergency Lockdown

One button to lock down your entire digital life if your phone/laptop is stolen or you suspect a breach.

- **Panic Button:** Single tap triggers pre-configured emergency actions across all connected accounts
- **Auto-Actions:** Revoke all active sessions, trigger password resets on critical accounts, enable logout on financial services
- **Trusted Contact Alert:** Auto-notify designated trusted contacts with instructions if you're unreachable
- **Recovery Plan:** Pre-built recovery sequence to regain access in the right order after lockdown

[X4] AI "Digital Twin" Risk Simulation

AI creates a virtual twin of your digital identity to stress-test it against emerging threats.

- **Synthetic Threat Scenarios:** AI generates novel attack scenarios (SIM swap + social engineering + credential stuffing combo attacks)
- **Vulnerability Discovery:** Finds attack paths you'd never think of — 'Your Spotify recovery email is the same as your bank, and Spotify was breached 3 times'
- **Continuous Red-Teaming:** Background AI agent periodically probes your digital twin for new weaknesses as your account network changes

[X5] Family/Team Privacy Shield

Protect your entire family or team's digital footprint from one dashboard.

- **Family Network View:** See how family members' accounts interconnect (shared Netflix, family iCloud, etc.)
- **Weakest Link Alert:** If one family member's security is weak, AI shows how it could compromise others
- **Parental Privacy Controls:** Monitor and manage children's app permissions and account exposure
- **Team Compliance (Enterprise):** For companies — ensure all employees meet security baselines, with ZKP-verified compliance reports

[X6] Data Broker Opt-Out Automation

AI finds where your data is being sold and auto-submits removal requests.

- **Broker Discovery:** AI scans known data broker databases (Spokeo, Whitepages, BeenVerified, etc.) for your personal info listings
- **Auto Opt-Out:** Automatically fills and submits opt-out/removal request forms on your behalf
- **Verification Follow-Up:** Re-checks after 30/60/90 days to confirm your data was actually removed
- **New Listing Alerts:** Continuous monitoring for new appearances of your data on broker sites

[X7] AR Privacy Scanner (Mobile Exclusive)

Point your camera at any app on your screen to see its privacy risk overlay in augmented reality.

- **Visual Risk Overlay:** Hold phone over laptop screen showing apps — AR overlay shows risk score, permissions, and breach status for each app icon
- **QR Privacy Check:** Scan any app's QR code to instantly see its privacy analysis before installing
- **Physical Device Audit:** Point at smart home devices (Alexa, Ring, etc.) to see what data they collect and their risk score

TECH STACK & ARCHITECTURE

FRONTEND (WEB)

React.js / Next.js + TailwindCSS + D3.js/Cytoscape.js for interactive graph visualization + Framer Motion for attack simulation animations

MOBILE APP

React Native / Flutter — cross-platform iOS & Android with native permission scanner APIs + biometric auth (FaceID/Fingerprint)

BACKEND

Node.js (Express/Fastify) or Python (FastAPI) — RESTful API + WebSocket for real-time breach alerts + Redis for caching

AI / ML ENGINE

Python — PyTorch Geometric for Graph Neural Networks (cascading risk) + HuggingFace Transformers for NLP policy analysis + scikit-learn for breach prediction + LangChain/Claude API for PrivacyBot chat

DATABASE

PostgreSQL (accounts, users) + Neo4j (account relationship graph) + MongoDB (unstructured breach data, policy documents)

BLOCKCHAIN

Polygon (L2 Ethereum) for gas-efficient on-chain hashing + Solidity smart contracts for audit trail + ZK-SNARKs (circom/snarkjs) for privacy proofs + IPFS for decentralized data storage

BREACH INTELLIGENCE

HaveIBeenPwned API + custom dark web scraper (Tor-based) + community DAO breach reporting

AUTH & SECURITY

OAuth 2.0 for inbox scanning + AES-256 encryption at rest + end-to-end encryption + DID/Verifiable Credentials (W3C standard)

INFRA & DEVOPS

Docker + Kubernetes + AWS/GCP + CI/CD (GitHub Actions) + Grafana/Prometheus monitoring

WEB vs. MOBILE APP — FEATURE MATRIX

Feature	Web App	Mobile App
Account Inventory & Import	Yes	Yes
Interactive Account Graph	Full (D3.js)	Simplified
App Permission Scanner	Browser Ext Only	Native API
AI Risk Scoring Engine	Yes	Yes
PrivacyBot Chat	Yes	Yes + Voice
Breach Alerts	Browser Push	Native Push
Attack Simulator	Full Animated	Simplified
Privacy Dashboard	Full Desktop	Mobile-Optimized
AR Privacy Scanner	No	Yes (Exclusive)
Panic Button / Lockdown	Yes	Yes + Widget
Biometric Auth	WebAuthn	FaceID / Fingerprint
NFT Badge Minting	Yes	Yes
ZKP Certificate	Yes	Yes
Offline Mode	No	Partial (cached data)
Dark Web Scan	Yes	Yes
Data Broker Opt-Out	Yes	Yes
Family Shield	Yes	Yes

UNIQUE SELLING POINTS

- **Not just a checker — a network analyzer:** Unlike HavelBeenPwned or password managers, we model cascading risk across your ENTIRE account graph
- **Blockchain-verified trust:** Your privacy posture is provably immutable — no one can fake or backdate a security audit
- **AI that thinks like a hacker:** Attack simulator and digital twin red-team YOUR accounts, not generic scenarios
- **Zero-Knowledge compliance:** Prove your security level without revealing anything — game-changer for insurance and enterprise
- **Cross-platform parity:** Full experience on web AND mobile, with mobile-exclusive features (AR scanner, native permissions, panic widget)

- **Community-powered intelligence:** Decentralized breach reporting DAO — faster, uncensorable, incentivized